

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

BIOLOGY — SSC-I (9th Grade)

Syllabus: Chapter 1 - The Science of Biology, Chapter 2 - Biodiversity, Chapter 3 - The Cell, Chapter 4 - Cell Cycle, Chapter 5 - Tissues, Organs and Organ Systems, Chapter 6 - Molecular Biology, Chapter 7 - Metabolism (Sections 7.1 to 7.6 only)

Total Marks: 43

Time Allowed: 2 Hours

Version 15

General Instructions:

1. Attempt **ALL** questions. **There are no choices in this paper.**
2. Section A carries 24 marks, Section B carries 9 marks and Section C carries 10 marks.
3. In Section A, fill the correct bubble (A, B, C or D) against each question. Cutting or overwriting a bubble will not be credited.
4. Write all working for Section B and Section C in the answer book. Marks are awarded for correct working, not only for the final answer.
5. Read every part of a question carefully and answer everything it asks for.
6. Use of a calculator is not allowed. Log and antilog tables may be used where required.

SECTION A — Multiple Choice Questions (24 × 1 = 24 Marks)

Attempt **ALL** questions. Fill the correct bubble against each question.

#	Question	A	B	C	D	Answer
1	The study of the development of an organism from a fertilized egg is called:	taxonomy	paleontology	embryology	histology	☐ A ☐ B ☐ C ☐ D
2	A statement that describes an observable occurrence in nature which appears to always be true is called a:	hypothesis	deduction	theory	scientific law	☐ A ☐ B ☐ C ☐ D
3	The collection of biological data through observations and experiments, and analysing it according to statistical rules, is called biostatistics. It is also known as:	biometry	bio-economics	biogeography	biophysics	☐ A ☐ B ☐ C ☐ D
4	A group of related genera makes up a:	class	family	order	phylum	☐ A ☐ B ☐ C ☐ D
5	The classification of living organisms into three domains is based on differences in the sequence of nucleotides in the:	DNA of the nucleus	cell wall proteins	rRNA of the cell	mRNA of the cytoplasm	☐ A ☐ B ☐ C ☐ D
6	A group of similar organisms whose individuals are capable of interbreeding and producing fertile offspring is called a:	species	genus	taxon	domain	☐ A ☐ B ☐ C ☐ D
7	According to the fluid mosaic model, the protein molecules of the cell membrane float like icebergs in a sea-like fluid of:	cellulose	cholesterol	carbohydrates	phospholipids	☐ A ☐ B ☐ C ☐ D
8	In a neuron, the cytoplasmic fibres that conduct nerve impulses <i>towards</i> the cell body are called:	axons	dendrites	cristae	flagella	☐ A ☐ B ☐ C ☐ D
9	The most basic stem cell, which has the ability to produce all the kinds of cell found in an organism, is the:	zygote	neuron	red blood cell	liver cell	☐ A ☐ B ☐ C ☐ D
10	The cell is regarded as the basic unit of life because it is the:	largest structure in the body	only structure that contains DNA	smallest unit of living material	only structure bounded by a membrane	☐ A ☐ B ☐ C ☐ D
11	In meiosis a diploid cell divides:	once, producing two haploid cells	once, producing four haploid cells	twice, producing two haploid cells	twice, producing four haploid cells	☐ A ☐ B ☐ C ☐ D

12	Which of the following takes place in meiosis but never in mitosis?	Formation of the spindle apparatus	Homologous chromosomes pair with each other	Chromatin condenses into chromosomes	The nuclear envelope disappears	A B C D
13	The replacement of worn-out cells and the healing of wounds depend on:	meiosis I	meiosis II	mitosis	crossing over	A B C D
14	In an animal cell, cytokinesis is completed by the inward growth of a:	cleavage furrow	phragmoplast	middle lamella	primary wall	A B C D
15	Which of the following is NOT one of the four main types of animal tissue?	Epithelial	Connective	Nervous	Vascular	A B C D
16	The inside of the stomach is lined by cells that secrete mucus, hydrochloric acid and the enzyme pepsin. These cells are:	muscle cells	epithelial cells	nerve cells	connective tissue cells	A B C D
17	The waxy substance that covers the upper epidermis of a leaf and forms the cuticle is:	cutin	lignin	pectin	chitin	A B C D
18	Every amino acid has a central α -carbon carrying a hydrogen atom, an amino group and a carboxyl group. One amino acid differs from another because of its:	amino group	carboxyl group	alkyl (R) group	hydrogen atom	A B C D
19	Lipids are insoluble in water but soluble in organic solvents such as alcohol and ether. Because of this they are described as:	crystalline	saturated	globular	hydrophobic	A B C D
20	Monosaccharides are named according to the number of carbon atoms they contain. A monosaccharide with six carbon atoms, such as glucose ($C_6H_{12}O_6$), is called a:	triose	hexose	pentose	tetrose	A B C D
21	The type of RNA that transfers amino acids to the ribosome during protein synthesis is:	tRNA	mRNA	rRNA	DNA	A B C D
22	The study of the structure, type and role of enzymes is called:	metabolism	physiology	enzymology	cytology	A B C D
23	The condensation of glucose molecules to form glycogen in muscle and liver cells is an example of:	digestion	anabolism	catabolism	inhibition	A B C D
24	ATP is converted to ADP and inorganic phosphate by the process of:	condensation	denaturation	inhibition	hydrolysis	A B C D

SECTION B — Short Questions (3 × 3 = 9 Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
2(i)	Table 2.1 of your textbook gives a simple classification of the pea plant, Pisum sativum . Write out its taxonomic ranks from Kingdom down to Species, giving the name your textbook lists at each rank.	3
2(ii)	An animal cell has no cell wall. State THREE advantages that an animal cell gains because it lacks a cell wall, as given in your textbook.	3
2(iii)	Table 6.1 of your textbook gives the location of each biomolecule inside the cell together with its main roles. For carbohydrates , lipids and DNA , state ONE location inside the cell and ONE main role of each.	3

SECTION C — Long Questions (2 × 5 = 10 Marks)

Attempt **ALL** questions. There are no choices.

Part	Question	Marks
3(i)	(a) Draw a well-labelled diagram of a nucleotide , showing its three components and stating to which carbon atom of the sugar each of the other two components is attached. [3] (b) Name the bond that holds the two strands of a DNA molecule together, state how many such bonds lie between adenine and thymine and how many lie between cytosine and guanine, and state what the backbone (uprights) of the DNA molecule is made of. [2]	5
3(ii)	A cell in the ovary of a plant contains 20 chromosomes and undergoes meiosis. (a) Name the TWO successive divisions of meiosis and state which one of them is the reduction division. [2] (b) Draw a flow chart showing how many cells are present, and how many chromosomes each cell contains, at the end of meiosis I and at the end of meiosis II. [2] (c) State whether the cells produced at the end of meiosis are haploid or diploid, and name the type of cells they will form. [1]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters covered: 1 The Science of Biology • 2 Biodiversity • 3 The Cell • 4 Cell Cycle • 5 Tissues, Organs and Organ Systems • 6 Molecular Biology • 7 Metabolism (7.1–7.6 only)

— END OF QUESTION PAPER —